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Third-Party Provider Signature Guidance

Learn how to create third-party payment provider signatures.

PSD2 signatures are intended for third-party provider (TPP) developers who make requests using their TPP application. An application is considered a TPP application if it contains a TPP role ([AISP](#) or [PISP](#)).

When TPP applications make calls to any TPP-related APIs in the EU region, a PSD2 signature is required. For calls that are related to Feeds and Reports APIs, you must add the signature when you interact with any TPP-specific Feed or Report type in the EU region.

To generate a successful signature, follow the instructions that are documented in the [RFC 9421](#) specification. Note that there are aspects of the signature generation that Amazon enforces for TPPs making SP-API requests. These are documented in the following sections.

Signature components

[RFC 9421](#) describes the process of communicating which HTTP message components are signed in a request. You must include the following message components when you generate a signature.

Note

Failure to include the following components results in a signature verification failure. You can add these components in any order.

HTTP fields (message headers)

Header	Description
x-amz-access-token	The LWA access token that is used to make the request.

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Header	Description
x-amzn-consistent-digest	The request payload's sha-256 digest. For requests without a payload, create the sha-256 digest on an empty string.

For additional information, refer to [HTTP Fields](#).

Derived components

Component	Description
@method	The request's HTTP method. The value must be upper case (for example <code>GET</code> , <code>POST</code>).
@query	The normalized query string that is defined by the URI and includes the leading <code>?</code> character. If the request doesn't have a query, you must populate the component value with <code>?</code> when you create the signature base mentioned in RFC 9421 . The value must be derived after you perform all the necessary encoding, such as Percentile-Encoding. Any further modifications you make to <code>queryString</code> , which are not included when you create the signature, cause signature failure.

For additional information, refer to [Derived Components](#).

Signature parameters

Parameter	Description
created	An epoch timestamp in seconds. If the <code>created</code> epoch points to a time older than five minutes by the time request is received, the signature is considered expired and the call is denied.
alg	An algorithm that you use to sign the signature base. The value for <code>alg</code> must be <code>PS512</code> . For additional details, refer to Signing algorithm .

For additional information, refer to [Signature Parameters](#).

 **Caution**

Special Authorization Types

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Third-Party Provider Signature Guidance

All request header values are case sensitive. If there is any case mismatch, your signature verification fails.

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Signature label

RFC 9421 allows multiple signatures within the same header (to disambiguate, use [signature labels](#)). SP-API only uses one signature, which is identified by the `x-amzn-psd2` signature label. You must use it in the `Signature` and `Signature-Input` headers to identify PSD2-related information.

Signing algorithm

A signature is generated on the signature base string that is created when you complete the [steps in RFC 9421](#). After you generate your signature base string, you must use the private key that is associated with your electronic NCA Certificate to sign the request with the [PS512](#) algorithm. This algorithm is the only one that is currently supported for SP-API requests.

The certificate that is associated with the private key you used in the signature must be passed as a header (`x-amzn-psd2-certificate`) in your SP-API request. The certificate must be a [PEM-formatted](#) string in Base64 encoding that begins with `-----BEGIN CERTIFICATE-----` and ends with `-----END CERTIFICATE-----`.

If you use any non-allowed characters in the certificate, your SP-API request is denied. You must use only certificates that are intended to sign HTTP requests that are issued by Qualified Trust Service Providers (QTSPs).

Delete an Application From Your Developer Account

CALLING THE SP-API

Configuration Details

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Caution

The certificate that you use must be associated with the organization identifier that you provided in your developer registration. If there is a mismatch, your request is denied.

Call Structure

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Error responses

If your request signature validation fails, you get a 403 error that contains additional information in the body. For example:

JSON

```
{
  "errors": [
    {
      "code": "Unauthorized",
      "message": "Access to requested
resource is denied.",

```

Performance Management

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Third-Party Provider Signature Guidance

```
    "details": "Reason for error."
  }
}
```

The following is a list of possible values for the details field:

Error message	Description
TPP certificate required but missing from request	The x-amzn-psd2-certificate header is missing.
TPP certificate has invalid format	The provided certificate is not in valid PEM format.
Content Digest header required but missing from request	The x-amzn-content-digest request header is missing.
Invalid Content Digest	The content digest couldn't be verified, or you used an unsupported algorithm. Only SHA-256 is supported.
Signature-Input header required but not presented	The Signature-Input request header is missing.
Signature-Input header is invalid	The Signature-Input header value doesn't adhere to RFC 9421 specification.
Request PSD2 Signature is Invalid	Invalid signature. This can have multiple causes, such as using the wrong private key or certificate, or a typo.
Signature header is required but not presented	The Signature request header is missing.

Example signature

The following example is a signature base string that contains all required components:

```
HTTP
"x-amz-access-token": Atza|IgEBIN...
"x-amzn-content-digest": sha-256=:fadf3qr3TAFASEG3234gADA+krdzMzYayg4YFig=:
"@method": POST
"@query": ?key2=value2&key1=value1
"@signature-params": ("x-amz-access-token" "x-
```

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```
amzn-content-digest" "@method"
"@query");created=1720137600;alg="PS512";
```

The associated `Signature-Input` header is:

```
HTTP

Signature-Input: x-amzn-psd2=("x-amz-access-
token" "x-amzn-content-digest" "@method"
"@query");created=1720137600;alg="PS512"
```

The following example is a signature header that has the `Signature` label:

```
HTTP

Signature: x-amzn-
psd2=:cyqStwls2k3Xs8SSzDvDhxWEc_P0dNxV5I6Tp...
:
```

 Updated 18 days ago

[← Third-Party Provider Registration](#) [Service Provider Registration and Role Approval Process Overview →](#)

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